

Supplementary Material for “Self-contrastive Learning to Boost Weakly Supervised Anomaly Detection”

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S-I. SUPPLEMENTARY METHOD DETAILS

A. Detailed Two-Stage Training Procedure

The complete two-stage training procedure of SEAD-Net is summarized in Algorithm S-I.

S-II. SUPPLEMENTARY EXPERIMENTAL DETAILS

A. Detailed Dataset Descriptions

As summarized in Tab. S-I, the NSL-KDD [1] dataset contains network traffic connection records described by 41 features, such as duration, service type, and source/destination information; the CelebA [2] dataset includes over 200,000 celebrity face images annotated with 39 attributes, where bald celebrities are treated as anomalies and the remaining samples (97%) are regarded as normal; the Census [3] dataset contains demographic and socioeconomic attributes and aims to identify high-income individuals (6%) earning over \$50,000 annually; the Fraud [4] dataset consists of credit card transactions with a small proportion of fraudulent cases; the Shuttle [5] and Mammography [5] datasets target anomaly detection in sensor observations and breast cancer diagnosis, respectively, with Shuttle providing nine sensor features; the Thyroid [6] dataset focuses on detecting thyroid diseases, particularly hypothyroidism; and the Bank [7] dataset contains customer information and marketing records from a Portuguese bank for predicting term deposit subscription.

B. Detailed Experimental Setup

Following the parameter settings commonly adopted in mainstream studies, the batch size is set to 512. The number of pre-training epochs is set to 200, and the number of fine-tuning epochs is set to 50. We employ the Adam optimizer to minimize the model loss function, with an initial learning

rate ranging from 0.01 to 0.001 and a weight decay fixed at 0.000001, while the model parameters remain unfrozen during training. To further enhance convergence stability and overall performance, a cosine annealing strategy is used to schedule the learning rate. In addition, 2% of anomalous samples are randomly selected and mixed into the normal class samples in each training dataset.

The experimental environment is configured as follows: Windows 11 Professional (64-bit) operating system, Intel Xeon Gold 5218 CPU @ 2.3GHz, 128GB DDR4 RAM, and NVIDIA GeForce RTX 5000 GPU with 16GB memory. The implementation is based on PyTorch 2.0.1, Python 3.8, and CUDA 11.7.

Algorithm S-I Two-Stage Network Training for SEAD-Net

Training stage 1: Pretraining

Input: All unlabeled data $\mathcal{X}^{\text{unlabeled}} \subset \mathbb{R}^M$ and $\mathcal{A}_e$ parameters ϕ_e

Output: $\Phi(\cdot; \phi_e)$ and $\mathbf{x}_{\text{cat}}$

- 1: Randomly initialize ϕ_e
- 2: **for** i in range($epoch_{1st}$) **do**
- 3: **for** j in range($\mathcal{X}^{\text{unlabeled}}_{\text{batch}}$) **do**
- 4: Compute the pretrain loss L_{1st}
- 5: Update parameters $\Phi(\cdot; \phi_e)$
- 6: **end for**
- 7: **end for**
- 8: **Return** $\Phi(\cdot; \phi_e)$ and $\mathbf{x}_{\text{cat}}$

Training stage 2: Finetuning

Input: All data $\mathcal{X} \subset \mathbb{R}^M$ and model parameters ϕ_e, ϕ_r and $\mathbf{x}_{\text{cat}}$

Output: $\Psi(\cdot; \phi_e, \phi_r)$ and y'

- 1: Randomly initialize ϕ_r and load $\phi_e, \mathbf{x}_{\text{cat}}$
 - 2: **for** i in range($epoch_{2st}$) **do**
 - 3: **for** j in range($\mathcal{X}_{\text{batch}}$) **do**
 - 4: Compute the finetune loss L_{2st}
 - 5: Update parameters $\Psi(\cdot; \phi_e, \phi_r)$
 - 6: **end for**
 - 7: **end for**
 - 8: **Return** $\Psi(\cdot; \phi_e, \phi_r)$ and y'
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TABLE S-I
DETAILED INFORMATION ABOUT VARIOUS DATASETS.

Datasets	Num	Dim	Train	Test	A_{no}	Domain
NSL-KDD	148,517	122	118814	29703	0.05%	Cybersecurity
Shuttle	49,097	9	39278	9819	0.08%	Aerospace
Fraud	284,807	29	227846	56961	0.01%	Finance
Bank	41,188	20	32950	8238	0.10%	Finance
Census	299,285	500	239428	59857	0.01%	Sociology
CelebA	202,599	39	162079	40520	0.02%	Cybersecurity
Thyroid	7,200	21	5760	1440	0.55%	Radiology
Mammography	11,183	6	8946	2237	0.34%	Radiology

S-III. ADDITIONAL EVALUATION METRICS

To further validate the advancement of SEAD-Net from a classification perspective, we evaluate its F1-score on four representative datasets and compare it with twelve competing methods, including TMG-GAN [8] and RODEO [9]. As shown in Tab. S-II, SEAD-Net achieves the highest average F1-score, which is consistent with the conclusions drawn from the AUC-ROC and AUC-PR metrics reported in the main manuscript.

Specifically, the average F1-score of SEAD-Net surpasses those of several representative state-of-the-art methods by 0.55% (Deep-SAD), 4.75% (TMG-GAN), 8.56% (RODEO), and 11.56% (Dual-MGAN), further confirming its strong detection capability and generalization performance under weak supervision.

TABLE S-II
COMPARISON OF THE F1 PERFORMANCE OF SEAD-Net WITH VARIOUS COMPETING METHODS.

Performances of F1-Score					
Datasets	Bank	CelebA	Fraud	Mammography	Avg.
Copod	0.4029	0.1512	0.0285	0.2850	0.2169
DevNet	0.3774	0.2178	0.2076	0.0917	0.2236
REPEN+	0.3661	0.2251	0.2123	0.1011	0.2262
FSNet+	0.3812	0.2195	0.2101	0.1012	0.2280
iForest	0.3573	0.2128	0.2321	0.1319	0.2335
Ecod	0.4085	0.1464	0.0925	0.2864	0.2335
Dual-MGAN	0.4720	0.3152	0.7143	0.2692	0.4427
RODEO	0.3868	0.2074	0.7598	0.5366	0.4727
FEWAE	0.5504	0.2860	0.6585	0.4571	0.4880
TMG-GAN	0.5402	0.3482	0.5170	0.6374	0.5107
Deep-SAD	0.5277	0.2993	0.7104	0.6735	0.5527
SEAD-Net	0.5802	0.3253	0.7489	0.5785	0.5582

S-IV. ADDITIONAL HYPERPARAMETER SENSITIVITY ANALYSIS

A. Impact of Different a_0 Values on SEAD-Net

This subsection evaluates the impact of the hyperparameter a_0 on SEAD-Net over the range $\{0.5, 1, 1.5, 2, 5, 10\}$, with other settings fixed, as shown in Tab. S-III. In general, SEAD-Net achieves its best performance when $a_0 = 1.5$, whereas setting $a_0 = 10$ degrades performance due to an excessively large fitting distance that hinders training. Across different a_0 values, SEAD-Net exhibits stable behavior, with average performance fluctuations of 1.95% (AUC-PR) and 0.61% (AUC-ROC); excluding the unstable Thyroid dataset, the fluctuations further decrease to 1.38% and 0.40%, respectively. Although

the Fraud dataset favors smaller a_0 values due to its specific characteristics, and the performance peaks for Census and Mammography appear at 1 and 2, these optima remain close to 1.5. Therefore, setting $a_0 = 1.5$ serves as an effective default choice for SEAD-Net.

TABLE S-III
COMPARISON OF THE PERFORMANCE OF SEAD-Net WITH DIFFERENT a_0 VALUES.

AUC-ROC						
Datasets	$a_0 = 0.5$	$a_0 = 1$	$a_0 = 1.5$	$a_0 = 2$	$a_0 = 5$	$a_0 = 10$
NSL-KDD	0.9800	0.9803	0.9832	0.9816	0.9830	0.9794
Shuttle	0.9974	0.9974	0.9975	0.9970	0.9943	0.9943
Fraud	0.9944	0.9942	0.9937	0.9933	0.9893	0.9939
Bank	0.9227	0.9220	0.9257	0.9226	0.9224	0.9218
Census	0.8877	0.8927	0.8913	0.8901	0.8883	0.8870
CelebA	0.9544	0.9561	0.9574	0.9563	0.9531	0.9514
Thyroid	0.9944	0.9940	0.9968	0.9838	0.9763	0.9838
Mammography	0.9766	0.9765	0.9769	0.9771	0.9767	0.9765

AUC-PR						
Datasets	$a_0 = 0.5$	$a_0 = 1$	$a_0 = 1.5$	$a_0 = 2$	$a_0 = 5$	$a_0 = 10$
NSL-KDD	0.9828	0.9814	0.9841	0.9827	0.9821	0.9818
Shuttle	0.9894	0.9900	0.9902	0.9890	0.9876	0.9878
Fraud	0.8176	0.8015	0.7979	0.7927	0.7894	0.7880
Bank	0.5732	0.5773	0.5819	0.5737	0.5734	0.5684
Census	0.3697	0.3832	0.3821	0.3762	0.3645	0.3616
CelebA	0.3224	0.3264	0.3291	0.3153	0.3172	0.3120
Thyroid	0.9309	0.9319	0.9528	0.9064	0.9138	0.8936
Mammography	0.7695	0.7703	0.7736	0.7705	0.7698	0.7639

B. Impact of Different λ Values on SEAD-Net

We evaluate SEAD-Net under different values of λ , varying between 0 and 1. When $\lambda = 1$, the optimization is dominated by the contrastive loss, whereas $\lambda = 0$ corresponds to a purely reconstruction-driven objective. As shown in Fig. S-I, SEAD-Net achieves its best performance when λ lies between 10^{-4} and 0.

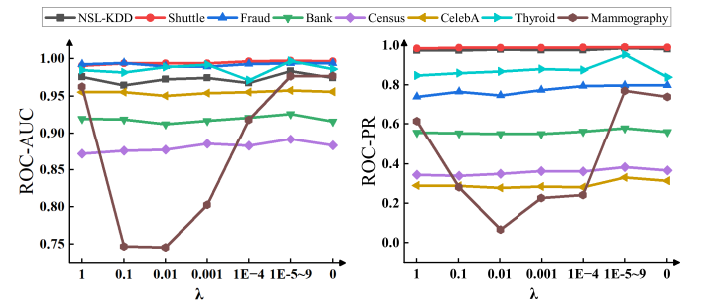


Fig. S-I. Comparison of the performance of SEAD-Net with different λ values in terms of AUC-ROC and AUC-PR.

For specific datasets such as NSL-KDD, Thyroid, and Mammography, increasing λ initially leads to a temporary performance drop, followed by a recovery to peak performance as λ decreases. This observation suggests that larger λ values constrain reconstruction optimization in the early stage. Notably, even with small λ , reinforcing the sample space remains critical for performance improvement, highlighting that careful tuning of λ enhances SEAD-Net on specific tasks rather than relying solely on a fixed model design.

S-V. ADDITIONAL ENHANCEMENT ANALYSIS

A. Performance of SEAD-Net with Combined Enhancement Methods

We conduct ablation experiments to evaluate the enhancement strategies introduced in the data enhancement module, comparing individual methods and selected combinations against the no-enhancement baseline. Here, the Performance Gain (PG) denotes the performance difference between each enhancement setting and the corresponding no-enhancement baseline. Fig. S-II reports the performance gains over the baseline.

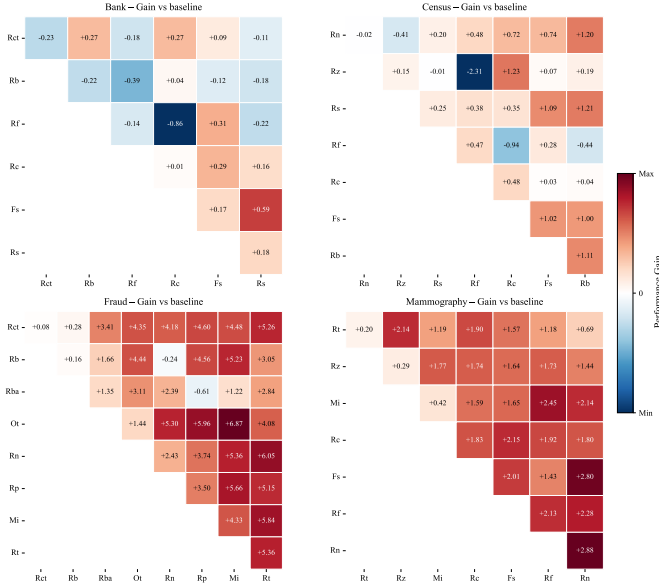


Fig. S-II. Performance gains of SEAD-Net under different enhancement methods and selected pairwise combinations.

On the Fraud dataset, eight enhancement methods, including *Random Contrast*, *Random BlockAug*, and *Opposite Time*, yield positive performance gains, with the combination of *Opposite Time* and *Mirroring* achieving a 6.87% improvement. However, some combinations of individually effective methods lead to slight performance degradation, indicating that excessive transformations may alter latent features and complicate learning. This effect is particularly evident on the Bank dataset, where most enhancements are ineffective except for *Random Cropping* and scale-related variants. In contrast, combining individually weaker methods, such as *Random Brightness* and *Random Contrast*, results in performance improvements. These results indicate that SEAD-Net effectively balances feature extraction and transformation-induced complexity, and exhibits clear performance advantages when sufficient additional features are captured to offset these challenges.

B. Visualization of the Latent Feature Space of SEAD-Net

To examine SEAD-Net’s representational capacity, we visualize its latent feature space using features from the intermediate hidden layer of the autoencoder and generate two-dimensional heatmaps. Four model variants are compared:

(i) BASE-Net, consisting solely of the reconstruction framework; (ii) SCLM-Net, integrating self-contrastive learning; (iii) DEM-Net, incorporating customized enhancement strategies; and (iv) SEAD-Net, combining both approaches. Fig. S-III presents the resulting latent feature visualizations.

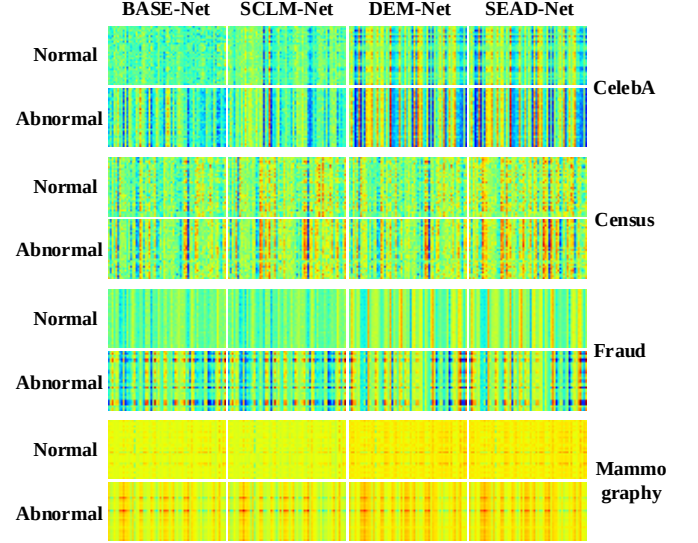


Fig. S-III. Visualization of the latent feature space of different model variants in SEAD-Net across multiple datasets.

As shown in Fig. S-III, the introduction of additional modules progressively enhances feature representations, as evidenced by increased contrast and sharper structural patterns. For instance, subtle differences between normal and anomalous samples in CelebA become more pronounced with added modules, while in Mammography, intensified patterns improve the separation of sample types. Comparable trends are observed across other datasets. By combining customized enhancement strategies and self-contrastive learning, SEAD-Net produces the most structured latent space, where normal and anomalous samples form well-separated and compact clusters, demonstrating its superior feature representation and robust anomaly discrimination.

REFERENCES

- [1] M. Tavallaei, E. Bagheri, W. Lu, and A. A. Ghorbani, “A detailed analysis of the KDD CUP 99 data set,” in *2009 IEEE Symposium on Computational Intelligence for Security and Defense Applications*, pp. 1–6, IEEE, 2009.
- [2] Z. Liu, P. Luo, X. Wang, and X. Tang, “Deep learning face attributes in the wild,” in *Proceedings of the IEEE International Conference on Computer Vision*, pp. 3730–3738, 2015.
- [3] “Census-Income (KDD).” UCI Machine Learning Repository, 2000. DOI: <https://doi.org/10.24432/C5N30T>.
- [4] X. Zhou, Z. Zhang, L. Wang, and P. Wang, “A model based on siamese neural network for online transaction fraud detection,” in *2019 International Joint Conference on Neural Networks (IJCNN)*, pp. 1–7, IEEE, 2019.
- [5] S. Rayana, “Statlog (Shuttle and Mammography).” UCI Machine Learning Repository, 2016. Available: <http://odds.cs.stonybrook.edu>.
- [6] R. Quinlan, “Thyroid Disease.” UCI Machine Learning Repository, 1987. DOI: <https://doi.org/10.24432/C5D010>.
- [7] S. Moro, P. Rita, and P. Cortez, “Bank Marketing.” UCI Machine Learning Repository, 2012. DOI: <https://doi.org/10.24432/C5K306>.

- [8] H. Ding, Y. Sun, N. Huang, Z. Shen, and X. Cui, "TMG-GAN: Generative adversarial networks-based imbalanced learning for network intrusion detection," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 1156–1167, 2023.
- [9] H. Mirzaei, M. Jafari, H. R. Dehbashi, A. Ansari, S. Ghobadi, M. Hadi, A. Soltani Moakhar, M. Azizmalayeri, M. S. Baghshah, and M. H. Rohban, "RODEO: Robust outlier detection via exposing adaptive out-of-distribution samples," in *Proceedings of the 41st International Conference on Machine Learning* (R. Salakhutdinov, Z. Kolter, K. Heller, A. Weller, N. Oliver, J. Scarlett, and F. Berkenkamp, eds.), vol. 235 of *Proceedings of Machine Learning Research*, pp. 35744–35778, PMLR, 21–27 Jul 2024.